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PHASE SHIFTER, ANTENNA, AND BASE STATION

Abstract

This application provides a phase shifter. The phase shifter includes a phase shift assembly including a substrate, a rotating arm, a drive arm, and a drive assembly. An output line is disposed on the substrate, and the output line includes an arc-shaped segment. The rotating arm is rotatably connected to the substrate through a rotating shaft, a coupling line is disposed on the rotating arm, and the coupling line is electrically connected to the arc-shaped segment. One end of the drive arm is fastened to the rotating arm, and the other end extends to a side that is of the rotating shaft and that faces away from the output line, and is in transmission connection to the drive assembly. The drive assembly drives the drive arm to move, and the drive arm drives the rotating arm to rotate relative to the substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/125728, filed on Oct. 20, 2023, which claims priority to Chinese Patent Application No. 202211447902.0, filed on Nov. 18, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This application relates to the field of communication technologies, and specifically, to a phase shifter, an antenna, and a base station.

BACKGROUND

[0003] An antenna of a base station implements beam downtilt adjustment through a phase shifter, and has many advantages such as a large downtilt angle adjustable range, high precision, an easily controlled directivity pattern, a strong anti-interference capability, and easy remote control. The phase shifter is one of core components of the antenna, and performance of the phase shifter directly affects overall performance of the antenna.

[0004] In a related technology, a problem of the phase shifter is that a relatively large force is required to drive a rotating arm to rotate, and consequently power costs of the phase shifter are relatively high, and it is not conducive to energy conservation.

SUMMARY

[0005] This application provides a phase shifter, an antenna, and a base station, to reduce a requirement of the phase shifter for power performance of a drive assembly, and alleviate a problem that a relatively large force is required to drive a rotating arm to rotate in a phase shifter in a related technology.

[0006] According to a first aspect, this application provides a phase shifter. The phase shifter includes a phase shift assembly. There may be one phase shift assembly, or may be a plurality of phase shift assemblies. When the phase shift assembly is specifically disposed, the phase shift assembly includes a substrate and a rotating arm, an output line is disposed on the substrate, and the output line includes an arc-shaped segment. A coupling line is disposed on the rotating arm, the rotating arm is rotatably disposed on the substrate through a rotating shaft, and the coupling line is electrically connected to the arc-shaped segment. The phase shift assembly further includes a drive arm and a drive assembly. One end of the drive arm is fastened to the rotating arm, and the other end of the drive arm extends to a side that is of the rotating shaft and that faces away from the output line, and is in transmission connection to the drive assembly. Specifically, the drive assembly drives the drive arm to swing, the drive arm drives the rotating arm to swing relative to the substrate, and in a process in which the rotating arm swings relative to the substrate, the coupling line slides along the arc-shaped segment of the output line. In this solution, the rotating arm is rotatably disposed on the substrate, and implements a transmission connection to the drive assembly through the drive arm extending to a side that is of a rotating shaft (the foregoing rotating shaft) and that faces away from the output line. A force arm acting on the rotating shaft is extended through the drive arm, so that the rotating arm can be driven, by using a relatively small force, to rotate, and the phase shifter has a lower requirement for power performance of the drive assembly.

[0007] When the fastened connection between the rotating arm and the drive arm is specifically implemented, in a technical solution, the rotating arm and the drive arm may be an integrated structure, to reduce a quantity of components of the phase shifter while connection reliability of the rotating arm and the drive arm is improved, thereby improving assembly efficiency and a lightweight degree of the phase shifter. In another technical solution, both the rotating arm and the drive arm may be independent components, and the two components are fastened to each other in a welding manner, a riveting manner, a bonding manner, or the like.

[0008] When the drive assembly is specifically disposed, the drive assembly may include a power output end, the power output end is in transmission connection to the drive arm, and the drive assembly drives the power output end to reciprocate along a first direction, so that the power output end drives the drive arm to drive the rotating arm to rotate, to implement a phase shift. In this solution, swing of the rotating arm is driven through rectilinear movement of the power output end, to help reduce a size of the phase shifter in a second direction. The second direction is a direction that is in a same plane in which the substrate is located as the first direction and that is perpendicular to the first direction.

[0009] In a specific technical solution, the first direction is in a same plane as a symmetry axis of the arc-shaped segment of the output line, and the first direction is perpendicular to the symmetry axis of the arc-shaped segment. In this solution, a movement path of the power output end is parallel to an edge of the substrate, so that the movement path of the power output end is shorter, thereby helping further improve miniaturization and a lightweight property of the phase shifter.

[0010] When the rotating shaft is specifically disposed, the rotating shaft may be located on the symmetry axis of the arc-shaped segment of the output line, so that a swing path of the rotating arm is symmetrical about the symmetry axis, thereby helping simplify a structural design of the output line, and design of a running parameter of the drive assembly, and the like.

[0011] When the transmission connection between the power output end and the drive arm is specifically implemented, in an optional technical solution, a guide trough extending along a length direction of the drive arm may be disposed on the drive arm, and the power output end includes a push shaft. The push shaft is inserted into the guide trough, and is driven by the drive assembly to push the drive arm and slide relative to the guide trough, to drive the drive arm to rotate. The transmission connection between the power output end and the drive arm is implemented through fitting between the push shaft and the guide trough. On one hand, a quantity of components is relatively small, and assembly efficiency is relatively high. On the other hand, fitting between the push shaft and the guide trough can further play a guiding role and a limiting role, to help ensure accuracy of a movement path of the rotating arm, and ensure reliable contact between the coupling line and the arc-shaped segment of the output line, thereby ensuring phase shift reliability.

[0012] In another optional technical solution, the phase shift assembly may include a sliding sleeve, and the sliding sleeve is sleeved onto the drive arm, and can slide relative to the drive arm along a length direction of the drive arm. In addition, an insertion hole is disposed in the sliding sleeve, and an axial direction of the insertion hole is parallel to an axial direction of the rotating shaft. A push shaft is inserted into the insertion hole, and can rotate relative to the insertion hole, to implement the transmission connection between the power output end and the drive arm through fitting between the push shaft and the sliding sleeve.

[0013] In a specific technical solution, the phase shift assembly may further include an elastic crimping assembly, and the elastic crimping assembly is fastened to the rotating arm, and clamps the substrate and the rotating arm, to apply an abutting force toward the substrate to the rotating arm. In this way, all gaps between the coupling line and the arc-shaped segment of the output line can be 0, thereby ensuring reliable contact between the coupling line and the arc-shaped segment of the output line, and maintaining a stable and reliable coupled connection relationship between the coupling line and the output line.

[0014] When the elastic crimping assembly is specifically disposed, the elastic crimping assembly

may include a crimping body, a crimping foot group, and a transition connecting part. The crimping foot group is located on a side that is of the substrate and that faces away from the rotating arm, and abuts against and is slidably connected to a surface that is of the substrate and that faces away from the rotating arm. The crimping body is fastened to a side that is of the rotating arm and that faces away from the substrate, and an elastic crimping part configured to press the rotating arm toward the substrate is disposed. The transition connecting part connects the crimping body and the crimping foot group. In this solution, the elastic crimping assembly is more conveniently disposed, and an effect of pressing the rotating arm toward the substrate by the elastic crimping assembly is more reliable.

[0015] In a specific technical solution, the substrate may have an arc-shaped edge, and the arc-shaped edge is located on a side that is of the output line and that faces away from the rotating shaft. The elastic crimping assembly slidably fits with the arc-shaped edge, to guide swing of the rotating arm through fitting between the elastic crimping assembly and the arc-shaped edge, so that the electrical connection between the coupling line and the arc-shaped segment of the output line is more reliable. The crimping foot group may include at least two crimping feet, the transition connecting part may include at least two connecting arms, and the crimping feet are spaced apart along an extension direction of the arc-shaped edge. The at least two connecting arms are connected to the at least two crimping feet in one-to-one correspondence, so that the elastic crimping assembly clamps the substrate and the rotating arm in the elastic crimping assembly. Space between two adjacent connecting arms may form a slot. An insertion block adapted to the slot is disposed at an end that is of the rotating arm and that is close to the arc-shaped edge, and the insertion block is inserted into the slot. This can implement a connection between the rotating arm and the elastic crimping assembly at this end through the insertion block and the slot, and also help improve a positioning speed and positioning accuracy between the elastic crimping assembly and the rotating arm during assembly of the phase shifter.

[0016] When the crimping body is specifically disposed, a hollow area may be disposed on the crimping body, and the elastic crimping part includes a spring plate disposed in the hollow area, to apply a force toward the substrate to the rotating arm through the spring plate. In this solution, the force toward the substrate can be applied to the rotating arm through a relatively small quantity of components, to help simplify a structure of the elastic crimping assembly, and also help improve crimping reliability.

[0017] In an optional technical solution, a quantity of hollow areas on the crimping body is the same as a quantity of output lines disposed on the substrate, and each output line is opposite to a spring plate in one hollow area. In this solution, each output line is opposite to a spring plate in one hollow area, and the spring plate applies an abutting force to the output line. This helps ensure a reliable connection and a 0 gap between an arc-shaped segment of each output line and the coupling line.

[0018] When the substrate is specifically disposed, two output lines may be disposed on the substrate, and the two output lines may be respectively denoted as a first output line and a second output line. A circle center of an arc-shaped segment of the first output line coincides with a circle center of an arc-shaped segment of the second output line, the first output line is farther away from the rotating shaft than the second output line, and both the arc-shaped segment of the first output line and the arc-shaped segment of the second output line are electrically connected to the coupling line. In this solution, the two output lines are disposed on the substrate, and each output line has two signal output ends, so that an antenna can transmit more signals.

[0019] In a specific technical solution, a main feeder and a coupling part are disposed on the substrate, both the main feeder and the coupling part are located on a side that is of the output line and that faces the rotating shaft, the main feeder is electrically connected to the coupling part, and an extension direction of the main feeder is parallel to the first direction, to help reduce a size occupied by the main feeder in the second direction, and facilitate miniaturization and a lightweight

property of the phase shifter.

[0020] A third output line may be further disposed on the substrate, to further increase a quantity of signals transmitted by the antenna. The third output line is located on the side that is of the output line and that faces the rotating shaft, and is electrically connected to the coupling part. An extension direction of the third output line is parallel to the first direction, to reduce a size occupied by the third output line in the second direction, and facilitate miniaturization and a lightweight property of the phase shifter.

[0021] In a specific technical solution, there are a plurality of phase shift assemblies, and the plurality of phase shift assemblies are sequentially disposed along the axial direction of the rotating shaft, to reduce a volume of the phase shifter.

[0022] When the phase shift assembly is specifically disposed, in an optional technical solution, the phase shift assembly includes a shielding panel, the substrate is fastened to the shielding panel, and the rotating arm is located on a side that is of the substrate and that faces away from the shielding panel. In this solution, the shielding panel can play a support role, to improve structural strength of the phase shift assembly, and can also reduce or even shield signal interference between two adjacent phase shift assemblies. One panel is used for two purposes, to help reduce a thickness of the phase shifter, thereby reducing space occupied by the phase shifter.

[0023] According to a second aspect, this application provides an antenna. The antenna includes a radiating element and the phase shifter in the first aspect. The radiating element is electrically connected to the phase shifter, and the phase shifter is configured to adjust a feeding phase of the radiating element. A requirement for power performance of the drive assembly in the phase shifter is relatively low, so that energy consumption of the antenna is lower.

[0024] According to a third aspect, this application provides a base station. The base station includes a mounting bracket and the antenna in the second aspect. The antenna is disposed on the mounting bracket, and is configured to receive or transmit a signal. The antenna has relatively low energy consumption, to help energy conservation of the base station.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 is a diagram of an application scenario of a base station according to a possible embodiment of this application;

[0026] FIG. 2 is a diagram of a structure of a base station according to a possible embodiment of this application;

[0027] FIG. 3 is a diagram of a structure of an antenna according to a possible embodiment of this application;

[0028] FIG. 4 is a front view of a phase shifter in a related technology;

[0029] FIG. 5 is a front view of a phase shifter according to a possible embodiment of this application;

[0030] FIG. 6 is an exploded view of the phase shifter shown in FIG. 5;

[0031] FIG. 7 is a diagram of a structure of a rotating arm and a drive arm in a phase shifter according to a possible embodiment of this application;

[0032] FIG. 8 is a diagram of a working principle of a phase shifter according to a possible embodiment of this application;

[0033] FIG. 9 is a diagram of a partial structure of a phase shifter according to a possible embodiment of this application;

[0034] FIG. 10 is a diagram of a connection relationship between a push shaft and a drive arm according to a possible embodiment of this application;

[0035] FIG. 11 is a diagram of a partial structure of a phase shifter according to another possible

embodiment of this application;

[0036] FIG. 12 is a diagram of a structure of an elastic crimping assembly of a phase shifter according to a possible embodiment of this application;

[0037] FIG. 13 is a diagram of a structure of an elastic crimping assembly, a rotating arm, and a substrate according to an embodiment of this application; and

[0038] FIG. 14 is a diagram of a structure of a phase shifter according to another possible embodiment of this application.

REFERENCE NUMERALS

[0039] **100**: antenna; **101**: radome; **102**: radiating element; **103**: reflection panel; **200**: mounting bracket; **300**: antenna adjustment bracket; **400**: radio frequency processing unit; **500**: baseband processing unit; **600**: cable; **700**: antenna connector; **810**: adjustment unit; **811**: transmission structure; **812**: calibration network; **820**: combiner; **830**: filter; and **900**: grounding device;

[0040] **1'**: phase shifter; **11'**: substrate; **12'**: rotating shaft; **13'**: rotating arm; and **14'**: drive assembly;

[0041] **1**: phase shifter; **10**: phase shift assembly; **10a**: first phase shift assembly; and **10b**: second phase shift assembly;

[0042] **11**: substrate; **111**: output line; **111a**: first output line; **111b**: second output line; **1111**: arc-shaped segment; **1112**: first output segment; **1113**: second output segment; **112**: rotating shaft mounting hole; **113**: main feeder; **114**: coupling part; and **115**: third output line;

[0043] **12**: rotating shaft; **13**: rotating arm; **131**: hole; **132**: coupling line; and **133**: insertion block;

[0044] **14**: drive assembly; **141**: power output end; **142**: push shaft; **1421**: shaft body; and **1422**: stopper;

[0045] **15**: drive arm; **151**: guide trough; **1511**: avoiding hole; and **16**: sliding sleeve;

[0046] **17**: elastic crimping assembly; **171**: crimping body; **1711**: through hole; **1712**: hollow area; **172**: crimping foot group; **1721**: crimping foot; **173**: transition connecting part; **1731**: connecting arm; **174**: elastic crimping part; **1741**: spring plate body; **1742**: crimping protrusion; **1743**: elastic member; **1744**: crimping member; and **1700**: slot; and

[0047] **18**: shielding panel; and **19**: metal housing.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0048] For ease of understanding a phase shifter, an antenna, and a base station provided in embodiments of this application, the following describes an application scenario of the phase shifter, the antenna, and the base station. FIG. 1 is a diagram of an example of an application scenario of a base station according to an embodiment of this application. As shown in FIG. 1, the application scenario may include a base station and a terminal. Wireless communication may be implemented between the base station and the terminal. The base station may be located in a base station subsystem (BSS), a UMTS terrestrial radio access network (UTRAN), or an evolved universal terrestrial radio access network (E-UTRAN), and is configured to perform cell coverage of a radio signal, to implement communication between a terminal device and a wireless network. Specifically, the base station may be a base transceiver station (BTS) in a global system for mobile communications (GSM) or a code division multiple access (CDMA) system, may be a NodeB (NB) in a wideband code division multiple access (WCDMA) system, may be an evolved NodeB (eNB or eNodeB) in a long term evolution (LTE) system, or may be a radio controller in a cloud radio access network (CRAN) scenario. Alternatively, the base station may be a relay station, an access point, a vehicle-mounted device, a wearable device, a g node (gNodeB or gNB) in a new radio (NR) system, a base station in a future evolved network, or the like. This is not limited in this embodiment of this application.

[0049] FIG. 2 is a diagram of a structure of a base station according to an embodiment of this application. The base station may usually include an antenna **100**, a mounting bracket **200**, an antenna adjustment bracket **300**, and other structures. The antenna **100** of the base station includes a radome **101**. The radome **101** has a good electromagnetic wave penetration property in terms of

electrical performance, and can withstand impact of an external harsh environment in terms of mechanical performance, so that the antenna **100** can be protected from impact of an external environment. The antenna **100** may be mounted on the mounting bracket **200** through the antenna adjustment bracket **300**, to facilitate receiving or transmitting of a signal of the antenna **100**.

[0050] In addition, the base station may further include a radio frequency processing unit **400** and a baseband processing unit **500**. For example, the radio frequency processing unit **400** may be configured to: perform frequency selection, amplification, and down-conversion processing on a signal received by the antenna **100**, convert the signal into an intermediate frequency signal or a baseband signal, and send the intermediate frequency signal or the baseband signal to the baseband processing unit **500**. Alternatively, the radio frequency processing unit **400** is configured to: perform up-conversion and amplification processing on a baseband signal or an intermediate frequency signal of the baseband processing unit **500**, convert the signal into an electromagnetic wave through the antenna **100**, and send the electromagnetic wave. The baseband processing unit **500** may be connected to a feeding network of the antenna **100** through the radio frequency processing unit **400**. In some implementations, the radio frequency processing unit **400** may also be referred to as a remote radio unit (RRU), or may be a radio frequency module in an active antenna unit (AAU). The baseband processing unit **500** may also be referred to as a baseband unit (BBU).

[0051] In a possible embodiment, as shown in FIG. 2, the radio frequency processing unit **400** and the antenna **100** may be integrally disposed, and the baseband processing unit **500** is located at a remote end of the antenna **100**. In some other embodiments, both the radio frequency processing unit **400** and the baseband processing unit **500** may be located at a remote end of the antenna **100**. The radio frequency processing unit **400** and the baseband processing unit **500** may be connected through a cable **600**.

[0052] More specifically, refer to FIG. 2 and FIG. 3 together. FIG. 3 is a diagram of a structure of an antenna according to a possible embodiment of this application. As shown in FIG. 3, the antenna **100** of the base station may include a radiating element **102** and a reflection panel **103**. The radiating element **102** may also be referred to as an antenna element, an element, or the like, and can effectively send or receive an antenna signal. In the antenna **100**, frequencies of different radiating elements **102** may be the same or different. The reflection panel **103** may also be referred to as a bottom plate, an antenna panel, a reflective surface, or the like, and may be made of metal. When the antenna **100** receives a signal, the reflection panel **103** may reflect the antenna signal to a target coverage area. When the antenna **100** transmits a signal, the reflection panel **103** may reflect and transmit the signal transmitted to the reflection panel **103**. The radiating element **102** is usually placed on a surface of one side of the reflection panel **103**. This can greatly enhance a signal receiving or transmitting capability of the antenna **100**, and can also block and shield interference caused to antenna signal receiving by another electromagnetic wave from a rear face of the reflection panel **103** (in this application, the rear face of the reflection panel **103** is a side opposite to the side that is of the reflection panel **103** and that is configured to dispose the radiating element **102**).

[0053] The antenna is connected to the radio frequency processing unit **400** through an antenna connector **700** located outside the radome **101**. In the antenna **100**, a feeding network is disposed between the radiating element **102** and the antenna connector **700**. The feeding network may provide specific power and a specific phase for the radiating element **102**. As shown in FIG. 3, the feeding network may include an adjustment unit **810** and a phase shifter **1**. The phase shifter **1** is configured to change a maximum direction of signal radiation. The phase shifter **1** correspondingly adjusts corresponding radiating elements **102**, so that electrical downtilt angles of radiation signals of the radiating elements **102** may be changed, to change radiation directions of the radiating elements **102**, thereby meeting a signal coverage requirement. The adjustment unit **810** is configured to implement different radiation beam directions, and may specifically include a transmission structure **811** and a calibration network **812**. The transmission structure **811** may drive

the phase shifter **1** to change directions of different radiation beams. The calibration network **812** sends a calibration signal to the transmission structure **811** to control an action of the transmission structure **811**.

[0054] Still refer to FIG. **3**. Some modules used for performance extension may be further disposed in the feeding network. For example, a combiner **820** may be further disposed. The combiner **820** may be configured to: combine signals of different frequencies into one signal, and transmit the signal through the antenna **100**; or during reverse use, the combiner **820** may be configured to: divide a signal received by the antenna **100** into a plurality of signals based on different frequencies, and transmit the signals to the baseband processing unit **500** for processing. For another example, a filter **830** may be further disposed. The filter **830** is configured to filter out an interference signal.

[0055] In a specific embodiment, a grounding device **900** may be disposed between the baseband processing unit **500** and the cable **600**, and the grounding device **900** usually includes a ground electrode buried underground. A sealing member may be disposed at a connecting place between the antenna **100** and the cable **600**, and a sealing member may also be disposed at a connecting place between the grounding device **900** and the cable **600**. The sealing member may specifically include at least one of an insulation sealing tape and a polyvinyl chloride (PVC) insulation tape. Certainly, the sealing member may alternatively be another structure, and is not limited to a tape form.

[0056] It is worth noting that any embodiment related to the term “specific”, such as “specifically”, “specifically disposed”, and “specifically designed” in this application is an optional embodiment. In other words, the embodiment is a possible specific embodiment under the inventive concept of this application, but other possible embodiments may be further included.

[0057] FIG. **4** is a front view of a phase shifter in a related technology. As shown in FIG. **4**, in the related technology, a phase shifter **1'** includes a substrate **11'**, a rotating shaft **12'**, a rotating arm **13'**, and a drive assembly **14'**. The rotating shaft **12'** is disposed on the substrate **11'**, and can rotate relative to the substrate **11'**. One end of the rotating arm **13'** is fastened to the rotating shaft **12'**. The drive assembly **14'** directly acts on the rotating shaft **12'** to drive the rotating shaft **12'** to rotate, to drive the rotating arm **13'** to rotate relative to the substrate **11'**, thereby implementing a phase shift. In this solution, a force driving the rotating shaft **12'** to rotate directly acts on the rotating shaft **12'**, and a force arm acting on the rotating shaft **12'** is almost zero. Torque is a product of the force and the force arm. To enable the rotating shaft **12'** to rotate, a relatively large force is required. As a result, the phase shifter **1'** has excessively large transmission resistance, the phase shifter **1'** has a higher requirement for power performance of the drive assembly **14'**, and energy consumption is higher. In addition, because the phase shifter **1'** has excessively large transmission resistance, the drive assembly has a relatively large transmission weight (the drive assembly has relatively large transmission resistance), and a failure rate of vibration joint adjustment is relatively high.

[0058] In view of this, embodiments of this application provide a phase shifter **1**, an antenna, and a base station, to alleviate the foregoing problem.

[0059] FIG. **5** is a front view of a phase shifter according to an embodiment of this application. FIG. **6** is an exploded view of the phase shifter shown in FIG. **5**. As shown in FIG. **5** and FIG. **6**, the phase shifter **1** includes a phase shift assembly **10**, and the phase shift assembly **10** includes a substrate **11** and a rotating arm **13**. Specifically, an output line **111** is disposed on the substrate **11**. The output line **111** may also be referred to as a strip, and the output line **111** and the substrate **11** jointly form a main printed circuit board (PCB). The output line **111** includes an arc-shaped segment **1111**, a first output segment **1112**, and a second output segment **1113**. For ease of description, one end of the arc-shaped segment **1111** is a first end, and the other end is a second end. The first output segment **1112** is electrically connected to the first end of the arc-shaped segment **1111**, and the second output segment **1113** is electrically connected to the second end of the arc-shaped segment **1111**. In a specific embodiment, the arc-shaped segment **1111**, the first

output segment **1112**, and the second output segment **1113** may be an integrated structure.

[0060] The rotating arm **13** is connected to the substrate **11**, and can rotate relative to the substrate **11**. A specific connection manner of the rotating arm **13** and the substrate **11** is not limited. For example, in a specific embodiment, a rotating shaft **12** is disposed on the substrate **11**, and the rotating shaft **12** can rotate relative to the substrate **11** under the action of an external force. Specifically, a rotating shaft mounting hole **112** may be disposed in the substrate **11**, and the rotating shaft **12** is inserted into the rotating shaft mounting hole **112** and can rotate relative to the rotating shaft mounting hole **112**. In another specific embodiment, a bearing mounting hole is disposed in the substrate **11**, a bearing is mounted in the rotating shaft mounting hole **112**, and the rotating shaft **12** is inserted into an inner hole of the bearing and is in interference fitting with the inner hole of the bearing. The rotating arm **13** is fastened to the rotating shaft **12**.

[0061] When fastening between the rotating arm **13** and the rotating shaft **12** is specifically implemented, the rotating shaft **12** and the rotating arm **13** may be an integrally formed structure, or fastened in another form. For example, a hole **131** is disposed in the rotating arm **13**, and the rotating shaft **12** is inserted into the hole **131** and is in interference fitting with the hole **131**; or a hole **131** is disposed in the rotating arm **13**, the rotating shaft **12** is inserted into the hole **131**, a limiting protrusion is disposed on one of the hole **131** and the rotating shaft **12**, a limiting groove matching the limiting protrusion is disposed on the other, and the limiting protrusion is inserted into the limiting groove, to implement fastening between the rotating arm **13** and the rotating shaft **12**.

[0062] When the rotating shaft **12** is specifically disposed, a material of the rotating shaft **12** is not limited, and the rotating shaft **12** may be an insulated rotating shaft, for example, a plastic rotating shaft. A coupling line **132** is disposed on the rotating arm **13**, and the coupling line **132** and the rotating arm **13** jointly form a coupling PCB. The coupling line **132** is connected to a coupling part **114** through coupling, and is also connected to the arc-shaped segment **1111** of the output line **111** through coupling. This is specifically described below. The rotating shaft **12** passing through the substrate **11** and the rotating arm **13** is an insulated rotating shaft, to help avoid a short circuit between the coupling part **114** and the coupling line **132**. Certainly, in another optional embodiment, the rotating shaft **12** may alternatively be a metal rotating shaft. In this case, it is only necessary to take an effective measure to avoid a short circuit. For example, at least one of the rotating arm **13** and the substrate **11** may be insulated from the rotating shaft **12**.

[0063] Still refer to FIG. 6. In a specific embodiment, a main feeder **113** and the coupling part **114** are disposed on the substrate **11**, and the main feeder **113** is electrically connected to the coupling part **114**. The coupling line **132** is disposed on the rotating arm **13**, one end of the coupling line **132** is electrically connected to the arc-shaped segment **1111** of the output line **111**, and the other end of the coupling line **132** is electrically connected to the coupling part **114**, to implement an electrical connection between the arc-shaped segment **1111** and the main feeder **113**.

[0064] In a specific embodiment, the coupling line **132** is connected to the arc-shaped segment **1111** of the output line **111** through coupling, and a first preset gap exists between the coupling line **132** and the arc-shaped segment **1111** of the output line **111**. The coupling line **132** is also connected to the coupling part **114** through coupling, and a second preset gap exists between the coupling line **132** and the coupling part **114**. A size of the first preset gap may be the same as or different from a size of the second preset gap. It is worth noting that the coupled connection may also be referred to as capacitive coupling. When two connection objects are connected through coupling, a signal can be transmitted between the two connection objects.

[0065] For example, a first insulation layer may be disposed between the coupling line **132** and the output line **111**, so that the first preset gap is formed between the coupling line **132** and the arc-shaped segment **1111**. Specifically, the first insulation layer may be disposed on the coupling line **132**, the first insulation layer may be disposed on the output line **111**, or the first insulation layer may be disposed on both the coupling line **132** and the output line **111**. A second insulation layer is

disposed between the coupling line **132** and the coupling part **114**, so that the second preset gap is formed between the coupling line **132** and the coupling part **114**. Specifically, the second insulation layer may be disposed on the coupling line **132**, the second insulation layer may be disposed on the coupling part **114**, or the second insulation layer may be disposed on both the coupling line **132** and the coupling part **114**.

[0066] Still refer to FIG. 5. The phase shift assembly **10** further includes a drive arm **15** and a drive assembly **14**. One end of the drive arm **15** is fastened to the rotating arm **13**, and the other end extends to a side that is of the rotating shaft **12** and that faces away from the output line **111**, and is in transmission connection to the drive assembly **14**. That the drive arm **15** is in transmission connection to the drive assembly **14** may be understood as follows: The drive arm **15** is connected to the drive assembly **14**, and the drive assembly **14** can drive the drive arm **15** to move.

Specifically, the drive assembly **14** can drive the drive arm **15** to swing, so that the drive arm **15** drives the rotating arm **13** to swing relative to the substrate **11**. In a process in which the rotating arm **13** swings relative to the substrate **11**, the coupling line **132** slides along the arc-shaped segment **1111** of the output line **111**, to implement a phase shift.

[0067] In the foregoing solution, in the phase shifter **1** of the antenna, the phase shift assembly **10** includes the drive arm **15**, the drive arm **15** extends to the side that is of the rotating shaft **12** and that faces away from the output line **111**, and the drive assembly **14** drives, through the drive arm **15**, the rotating arm **13** to rotate, to implement a phase shift. A force arm acting on the rotating shaft **12** is extended because the drive arm **15** is disposed. Compared with a related technology, only a smaller force is required to drive the rotating arm **13** to rotate. This can reduce a requirement for power performance of the drive assembly **14**, reduce design complexity of the phase shifter **1**, and reduce energy consumption and costs; and can also effectively alleviate a problem that the drive assembly has a relatively large transmission weight and a failure rate of vibration joint adjustment is relatively high due to excessively large transmission resistance of the phase shifter **1**.

[0068] When the fastened connection between the drive arm **15** and the rotating arm **13** is specifically implemented, a specific connection manner of the drive arm **15** and the rotating arm **13** is not limited, provided that the fastened connection between the drive arm **15** and the rotating arm **13** can be implemented.

[0069] FIG. 7 is a diagram of a structure of a rotating arm and a drive arm according to an embodiment of this application. As shown in FIG. 7, in an optional embodiment, the rotating arm **13** and the drive arm **15** are an integrated structure. The rotating arm **13** and the drive arm **15** use an integrated structure, so that connection strength of the rotating arm **13** and the drive arm **15** can be improved, and an assembly process of the rotating arm **13** and the drive arm **15** can be further omitted, to help reduce an assembly error, thereby improving precision of the phase shifter **1**.

[0070] Certainly, the rotating arm **13** and the drive arm **15** may alternatively be a split structure, in other words, both the rotating arm **13** and the drive arm **15** are independent components. In this case, the rotating arm **13** may be fastened to the drive arm **15** in a welding manner, a riveting manner, a bolt connection manner, or the like.

[0071] When the substrate **11** is specifically disposed, a quantity of output lines **111** may be set based on an actual requirement. For example, one output line **111** may be disposed, or two output lines **111** may be disposed on the substrate **11**. The following describes a working principle of the phase shifter **1** by using an example in which two output lines **111** are disposed on the substrate **11**.

[0072] For ease of description, in the two output lines **111**, one output line **111** far away from the rotating shaft **12** is denoted as a first output line **111a**, and the other output line **111** is denoted as a second output line **111b**. The first output line **111a** and the second output line **111b** each include an arc-shaped segment **1111**, a first output segment **1112**, and a second output segment **1113**, circle centers of arc-shaped segments **1111** of the two output lines **111** coincide, and both the arc-shaped segments **1111** are electrically connected to the coupling line **132**.

[0073] FIG. 8 is a diagram of a working principle of a phase shifter according to an embodiment of

this application. Specifically, as shown in FIG. 8, a first output segment **1112** of the first output line **111a** is configured to output a first signal, and an end that is of the first output segment **1112** and that is far away from an arc-shaped segment **1111** may be denoted as a first signal output end **M1**. A second output segment **1113** of the first output line **111a** is configured to output a second signal, and an end that is of the second output segment **1113** and that is far away from the arc-shaped segment **1111** may be denoted as a second signal output end **M2**. A first output segment **1112** of the second output line **111b** is configured to output a third signal, and an end that is of the first output segment **1112** and that is far away from an arc-shaped segment **1111** may be denoted as a third signal output end **M3**. A second output segment **1113** of the second output line **111b** is configured to output a fourth signal, and an end that is of the second output segment **1113** and that is far away from the arc-shaped segment **1111** may be denoted as a fourth signal output end **M4**.

[0074] One end of the main feeder **113** is electrically connected to the coupling part **114**, and the other end of the main feeder **113** is a signal input end **M0**. The signal input end is configured to receive an electrical signal input to the phase shifter **1**, and the electrical signal may be denoted as an input electrical signal.

[0075] During working, the input electrical signal enters from the signal input end of the main feeder **113**, is transferred to the two output lines **111** through the main feeder **113**, the coupling part **114**, and the coupling line **132**, and is divided into two signals, namely, left and right signals, from an electrical connection place between the coupling line **132** and each output line **111**. In the first output line **111a**, one electrical signal passes through a partial output line **111** between the first signal output end **M1** and the coupling line **132** and is output from the first signal output end **M1**, and the other electrical signal passes through a partial output line **111** between the second signal output end **M2** and the coupling line **132** and is output from the second signal output end **M2**. In the second output line **111b**, one electrical signal passes through a partial output line **111** between the third signal output end **M3** and the coupling line **132** and is output from the third signal output end **M3**, and the other electrical signal passes through a partial output line **111** between the fourth signal output end **M4** and the coupling line **132** and is output from the fourth signal output end **M4**. The coupling line **132** is coupled to the first output line **111a** and the second output line **111b** in a process of swinging left and right, to implement a change of a physical length of the output line **111**, to implement a change of a phase of each signal output end and implement a phase shift.

[0076] In a specific embodiment, the phase shifter **1** uses a disc phase shift structure, and implements a phase shift function based on a strip physical phase shift scenario by controlling a relative position of the coupling line **132** to the output line **111**.

[0077] Still refer to FIG. 5. When the drive assembly **14** is specifically disposed, the drive assembly **14** includes a power output end **141**, and the power output end **141** is in transmission connection to the drive arm **15**. In an embodiment, the drive assembly **14** drives the power output end **141** to reciprocate along a first direction **A**, and the power output end **141** drives the drive arm **15** to swing, to drive the rotating arm **13** to rotate, to implement a phase shift. For example, the drive assembly **14** may be a metal structure, or may be a non-metal structure. This is not limited in this embodiment of this application.

[0078] In an optional technical solution, the first direction **A** is in a same plane as a symmetry axis of the arc-shaped segment **1111**, and is perpendicular to the symmetry axis of the arc-shaped segment **1111**.

[0079] Usually, an edge of an end that is of the substrate **11** and that is far away from the output line **111** is a straight line, so that the direction in which the power output end **141** reciprocates is perpendicular to the symmetry axis of the arc-shaped segment **1111**, in other words, a movement direction of the power output end **141** is parallel to the edge of the substrate **11**, so that a movement path of the power output end **141** is shorter. This helps reduce design difficulty of a structure of the arc-shaped segment **1111**, a movement speed of the power output end **141**, and the like, and also helps reduce a size occupied by the phase shifter **1** in a direction of the symmetry axis of the arc-

shaped segment **1111**, thereby facilitating miniaturization and a lightweight property of the phase shifter **1**.

[0080] It is worth noting that, in this embodiment of this application, “perpendicular” is described for a current process level, but is not an absolute strict definition in a mathematical sense. A deviation of a predetermined angle may exist between the first direction A and the symmetry axis of the arc-shaped segment **1111**. In other words, an included angle between the first direction and the symmetry axis of the arc-shaped segment **1111** is not necessarily strict 90° , and may be 85° , 86° , 87° , 88° , 89° , 91° , 92° , 93° , 94° , 95° , or the like. In this embodiment of this application, “parallel” is described for a current process level, but is not an absolute strict definition in a mathematical sense. A deviation of a predetermined angle may exist between the movement direction of the power output end **141** and the edge of the substrate **11**. In other words, an included angle between the movement direction of the power output end **141** and the edge of the substrate **11** is not necessarily strict 0° or 180° . For example, an included acute angle between the movement direction of the power output end **141** and the edge of the substrate **11** may be 0.5° , 1° , 1.5° , 2° , 3° , 4° , 4.5° , 5° , or the like.

[0081] In an embodiment, the rotating shaft **12** is located on the symmetry axis of the arc-shaped segment **1111**, so that a movement track of the rotating arm **13** is symmetrical about the symmetry axis of the arc-shaped segment **1111**, and structural designs of the drive assembly **14** and the output line **111** are more simplified.

[0082] In another embodiment, the rotating shaft **12** may alternatively be located outside the symmetry axis of the arc-shaped segment **1111**, and details are not described in this implementation.

[0083] When the main feeder **113** and the coupling part **114** are specifically disposed on the substrate **11**, in an embodiment, both the main feeder **113** and the coupling part **114** are located on a side that is of the output line **111** and that faces the rotating shaft, and an extension direction of the main feeder **113** is parallel to the first direction A.

[0084] An included angle between the extension direction of the main feeder **113** and the first direction A is 0° , so that a size occupied by the main feeder **113** in a second direction B is smaller, to help reduce a size of the phase shift assembly **10** in the second direction B, and facilitate miniaturization and a lightweight property of the phase shifter **1**. Both the second direction B and the first direction A are in a plane in which the substrate **11** is located, and the second direction B is perpendicular to the first direction A.

[0085] For example, the coupling part **114** may be in a ring shape, and the rotating shaft **12** passes through a ring hole in the ring shape.

[0086] Still refer to FIG. 8. Further, the phase shifter **1** may further include a third output line **115** disposed on the substrate **11**. The third output line **115** is located on the side that is of the output line **111** and that faces the rotating shaft **12**, and is electrically connected to the coupling part **114**. An end that is of the third output line **115** and that is far away from the coupling part **114** may be denoted as a fifth output end M5, and the fifth output end M5 is configured to transmit an electrical signal, so that a quantity of signals transmitted by the antenna can be increased.

[0087] In a specific embodiment, an extension direction of the third output line **115** is parallel to the first direction A, so that a size occupied by the third output line **115** in the second direction B is smaller, to help reduce a size of the phase shift assembly **10** in the second direction B, and facilitate miniaturization and a lightweight property of the phase shifter **1**.

[0088] When the third output line **115** is specifically disposed, in an embodiment, the third output line **115** and the main feeder **113** may be located on a same side of the coupling part **114** in the first direction A. In another embodiment, the third output line **115** and the main feeder **113** may alternatively be located on different sides of the coupling part **114** in the first direction. For example, the main feeder **113**, the coupling part **114**, and the third output line **115** may be sequentially disposed along the first direction A.

[0089] When the arc-shaped segment **1111** of the output line **111** is specifically disposed, the arc-shaped segment **1111** may be a smooth arc-shaped line, or may include several V-shaped, “s”-shaped, “n”-shaped, “Ω”-shaped, “M”-shaped, or “W”-shaped substructures connected end to end, where the substructures are connected in an arc-shaped direction.

[0090] Still refer to FIG. **8**. In an embodiment, a guide trough **151** is disposed on the drive arm **15**, and the guide trough **151** extends along a length direction of the drive arm **15**. The power output end **141** includes a push shaft **142**, and the push shaft **142** is inserted into the guide trough **151**, and can be driven by the drive assembly **14** to slide relative to the guide trough **151**. In this solution, the guide trough **151** can be disposed to guide and limit movement of the push shaft **142**, to ensure a movement path of the rotating arm **13**, and ensure reliable contact between the coupling line **132** and the arc-shaped segment **1111** of the output line **111**, thereby ensuring phase shift reliability.

[0091] It is worth noting that, in this embodiment of this application, a length direction of an object X may be understood as follows: The object X extends along approximately two directions in a plane in which the object X is located: a third direction and a fourth direction, where the third direction and the fourth direction are perpendicular to each other. A size of the object X in the third direction is C, and a size of the object X in the fourth direction is D. If $C > D$, the third direction is the length direction of the object X, and the fourth direction is a width direction of the object X.

[0092] FIG. **9** is a diagram of a connection relationship between a push shaft and a drive arm according to an embodiment of this application. As shown in FIG. **9**, when the push shaft **142** is specifically disposed, the push shaft **142** may include a shaft body **1421** and a stopper **1422**, and an axial direction of the shaft body **1421** is parallel to an axial direction of the rotating shaft **12**. Specifically, the drive assembly **14** includes a power part, and the power part is in transmission connection to the push shaft **142**, to drive the push shaft **142** to move along the first direction A. One end of the shaft body **1421** is in transmission connection to the power part, and the other end is fastened to the stopper **1422** after passing through the guide trough **151**.

[0093] When the push shaft **142** is specifically disposed, a diameter of the shaft body **1421** is slightly less than a width of the guide trough **151**, so that the shaft body **1421** is in clearance fitting with the guide trough **151**. An outline size of the stopper **1422** is greater than the width of the guide trough **151**, to implement axial limiting of the shaft body **1421**.

[0094] Further, in an optional embodiment, the stopper **1422** and the shaft body **1421** are an integrated structure, so that a connection between the stopper **1422** and the shaft body **1421** is more reliable. Correspondingly, in this case, an avoiding hole **1511** of the stopper **1422** is disposed at an end that is of the guide trough **151** and that is close to the rotating shaft **12**, so that the stopper **1422** can pass through the avoiding hole **1511**, and the shaft body **1421** can slide in the guide trough **151**, so that the stopper **1422** plays an axial limiting role for the shaft body **1421**.

[0095] Certainly, a connection manner of the stopper **1422** and the shaft body **1421** is not limited to the integrated structure. For example, both the stopper **1422** and the shaft body **1421** may alternatively be independent components, and the two components are fastened. Further, the two components may be disassembled.

[0096] A specific structure of the stopper **1422** is not limited, and an outline of the avoiding hole **1511** in the guide trough **151** may match an outline of the stopper **1422**. For example, in an embodiment, the stopper **1422** is plate-shaped. Specifically, the stopper **1422** may be circular plate-shaped, square plate-shaped, or the like. In another embodiment, the stopper **1422** may be rod-shaped or block-shaped.

[0097] FIG. **10** is a diagram of another connection relationship between a push shaft and a drive arm according to an embodiment of this application. As shown in FIG. **10**, when the connection between the drive arm **15** and the power output end **141** is specifically implemented, in another embodiment, the phase shift assembly **10** may include a sliding sleeve **16**, and the sliding sleeve **16** is sleeved onto the drive arm **15**, and can slide relative to the drive arm **15** along a length direction of the drive arm **15**. An insertion hole is disposed in the sliding sleeve **16**, and an axial direction of

the insertion hole is parallel to an axial direction of the rotating shaft **12**. A push shaft **142** is inserted into the insertion hole, and can rotate relative to the insertion hole.

[0098] FIG. **11** is a diagram of a partial structure of a phase shifter according to an embodiment of this application. As shown in FIG. **11**, in a specific embodiment, the phase shift assembly **10** may include an elastic crimping assembly **17**. The elastic crimping assembly **17** is fastened to the rotating arm **13**, and clamps the substrate **11** and the rotating arm **13**, to apply an abutting force toward the substrate **11** to the rotating arm **13**, so that all gaps between the coupling line **132** and the arc-shaped segment **1111** of the output line **111** are 0, thereby implementing reliable contact between the coupling line **132** and the arc-shaped segment **1111** of the output line **111**, and maintaining a stable coupled connection relationship between the coupling line **132** and the output line **111**.

[0099] Uniformity of the gaps between the coupling line **132** and the arc-shaped segment **1111** of the output line **111** has relatively large impact on standing wave consistency of the antenna. Smaller and more uniform gaps indicate better standing wave consistency of the antenna. In this embodiment, the elastic crimping assembly **17** applies the abutting force toward the substrate **11** to the rotating arm **13**, so that all the gaps between the coupling line **132** and the arc-shaped segment **1111** of the output line **111** are 0, and all the gaps between the coupling line **132** and the arc-shaped segment **1111** of the output line **111** are 0. This meets a “small” gap requirement, and also achieves better uniformity. In this way, feeding performance, consistency, and stability of the phase shifter **1** provided in this embodiment are more stable, and standing wave consistency of the antenna is better. In addition, the rotating arm **13** can ensure a proper push-pull force.

[0100] It should be noted that the elastic crimping assembly **17** has relatively small impact on a push-pull force of the phase shifter **1**, and has relatively small impact on a transmission weight pressure of the phase shifter **1**.

[0101] In a specific embodiment, to avoid a short circuit, the elastic crimping assembly **17** may have an insulation property. For example, the elastic crimping assembly **17** may be an elastic plastic member, another insulated elastic non-metal member, or an elastic metal member on which an insulation layer is disposed. In an optional embodiment, the elastic crimping assembly **17** is an elastic plastic member formed through integral injection molding, to achieve a better crimping effect while avoiding a short circuit.

[0102] FIG. **12** is a diagram of a structure of an elastic crimping assembly according to an embodiment of this application. As shown in FIG. **11** and FIG. **12**, in a specific embodiment, the elastic crimping assembly **17** may include a crimping body **171**, a crimping foot group **172**, and a transition connecting part **173**. The crimping foot group **172** is located on a side that is of the substrate **11** and that faces away from the rotating arm **13**, and abuts against and is slidably connected to a surface that is of the substrate **11** and that faces away from the rotating arm **13**. The crimping body **171** is fastened to a side that is of the rotating arm **13** and that faces away from the substrate **11**, and an elastic crimping part **174** configured to press the rotating arm **13** toward the substrate **11** is disposed. The transition connecting part **173** connects the crimping body **171** and the crimping foot group **172**, so that the crimping body **171** and the crimping foot group **172** can jointly act to clamp the substrate **11** and the rotating arm **13**.

[0103] In an embodiment, the substrate **11** may have an arc-shaped edge, and the arc-shaped edge is located on a side that is of the output line **111** and that faces away from the rotating shaft **12**. The elastic crimping assembly **17** slidably fits with the arc-shaped edge. In a specific embodiment, a circle center of the arc-shaped edge coincides with a circle center of the arc-shaped segment **1111** of the output line **111**, to play a guiding role for movement of the rotating arm **13** through fitting between the elastic crimping assembly **17** and the arc-shaped edge, so that contact between the coupling line **132** and the arc-shaped segment **1111** is more reliable.

[0104] The crimping foot group **172** includes at least two crimping feet **1721**, the transition connecting part **173** includes at least two connecting arms **1731**, and the crimping feet **1721** are

spaced apart along an extension direction of the arc-shaped edge. The connecting arms **1731** are connected to the crimping feet **1721** in one-to-one correspondence, and space between two adjacent connecting arms **1731** forms a slot **1700**. An insertion block **133** adapted to the slot **1700** is disposed at an end that is of the rotating arm **13** and that is close to the arc-shaped edge, and the insertion block **133** is inserted into the slot **1700**, to implement an end connection between the rotating arm **13** and the elastic crimping assembly **17**. In a process in which the rotating arm **13** swings, the insertion block **133** and the slot **1700** form a tangential force, so that the rotating arm **13** drives the elastic crimping assembly **17** to synchronously move. In addition, fitting between the slot **1700** and the insertion block **133** also facilitates quick positioning between the elastic crimping assembly **17** and the rotating arm **13**.

[0105] In this embodiment, a correspondence between the insertion block **133** and the slot **1700** is not limited. For example, insertion blocks **133** may be in one-to-one correspondence with slots **1700**, and each insertion block **133** is inserted into a corresponding slot **1700**. Alternatively, a quantity of insertion blocks **133** may be less than a quantity of slots **1700**, but each insertion block **133** matches a slot **1700**.

[0106] In an optional embodiment, the crimping foot group **172** includes two crimping feet **1721** spaced apart along the extension direction of the arc-shaped edge, and space between the two crimping feet **1721** forms a slot **1700**. An insertion block **133** is disposed at the end that is of the rotating arm **13** and that is close to the arc-shaped edge, and the insertion block **133** is inserted into the slot **1700**. In this solution, the crimping foot group **172** includes only two crimping feet **1721**, so that the elastic crimping assembly **17** has a simpler structure and a smaller size, thereby facilitating miniaturization and a lightweight property of the phase shifter **1**.

[0107] To make the connection between the elastic crimping assembly **17** and the rotating arm **13** more reliable, in an embodiment, the crimping body **171** is fastened to the rotating shaft **12**. For example, a through hole **1711** is disposed at an end that is of the crimping body **171** and that is close to the rotating shaft **12**, and the rotating shaft **12** is in interference fitting with the through hole **1711**, or the end that is of the crimping body **171** and that is close to the rotating shaft **12** is welded onto the rotating shaft **12**.

[0108] As shown in FIG. **12**, when the elastic crimping part **174** is specifically disposed, in an optional embodiment, a hollow area **1712** may be formed on the crimping body **171**, the elastic crimping part **174** includes a spring plate, and the spring plate is disposed in the hollow area **1712**. In this solution, the elastic crimping part **174** has a relatively small quantity of components, so that crimping is more accurate and reliable. In addition, a structure of the elastic crimping assembly **17** can also be simplified.

[0109] In this embodiment, a shape of the hollow area **1712**, a connection position of the spring plate in the hollow area **1712**, and a quantity of spring plates are not limited, provided that the spring plate can apply an abutting force to the rotating arm **13** to press the rotating arm **13** toward the substrate **11**. For example, the hollow area **1712** may be circular, elliptical, rectangular, or the like, and there may be one spring plate, two spring plates, three spring plates, or the like in the hollow area **1712**.

[0110] For example, the hollow area **1712** is rectangular. When one spring plate is disposed in the hollow area **1712**, the spring plate may be disposed on any side of the hollow area **1712**. When a plurality of spring plates are disposed in the hollow area **1712**, positions of the spring plates in the hollow area **1712** may be set based on an actual requirement. For example, the plurality of spring plates may be evenly distributed in the hollow area **1712**, so that the rotating arm **13** is more evenly stressed. It is worth noting that, “a plurality of” mentioned in this embodiment means “greater than or equal to 2”.

[0111] In addition, in this embodiment, a quantity of hollow areas **1712** on the crimping body **171** is not limited. For example, one hollow area **1712**, two hollow areas **1712**, or three hollow areas **1712** may be disposed on the crimping body **171**. When the quantity of hollow areas **1712** on the

crimping body **171** is greater than or equal to 2, an arrangement of hollow areas **1712** on the crimping body **171** may be set based on an actual requirement.

[0112] In a specific embodiment, a quantity of hollow areas **1712** on the crimping body **171** is the same as a quantity of output lines **111** disposed on the substrate **11**, and each output line **111** is opposite to a spring plate in one hollow area **1712**. For example, when two output lines **111**, namely, the first output line **111a** and the second output line **111b**, are disposed on the substrate **11**, two hollow areas **1712** are disposed on the crimping body **171**, the two hollow areas **1712** are arranged along a length direction of the crimping body **171**, at least one spring plate is disposed in each hollow area **1712**, and each spring plate is connected to an end that is of a corresponding hollow area **1712** and that is close to the rotating shaft **12**. In addition, each spring plate in one hollow area **1712** is opposite to the arc-shaped segment **1111** of the first output line **111a**, and each spring plate in the other hollow area **1712** is opposite to the arc-shaped segment **1111** of the second output line **111b**.

[0113] In this embodiment, a quantity of elastic crimping assemblies **17** is not limited. For example, there may be one elastic crimping assembly **17**, two elastic crimping assemblies **17**, or three elastic crimping assemblies **17**. When the quantity of elastic crimping assemblies **17** is greater than or equal to 2, each elastic crimping assembly **17** is fastened to the rotating arm **13**, to synchronously rotate with the rotating arm **13**, and always press the rotating arm **13** toward the substrate **11**.

[0114] A specific shape of the spring plate is not limited. For example, the spring plate may be a rectangular, trapezoidal, or circular planar structure, and may further include a spring plate body **1741** and a crimping protrusion **1742**. The spring plate body **1741** is a rectangular, trapezoidal, or circular planar structure, and the crimping protrusion **1742** is disposed on the spring plate body **1741** and is located on a side that is of the spring plate body **1741** and that faces the rotating arm **13**.

[0115] FIG. **13** is a diagram of a structure of an elastic crimping assembly, a rotating arm, and a substrate according to an embodiment of this application. As shown in FIG. **13**, in another embodiment, the elastic crimping part **174** may alternatively use another structure. For example, the elastic crimping part **174** may include an elastic member **1743** and a crimping member **1744**. The crimping member **1744** is located between the crimping body **171** and the rotating arm **13**, the elastic member **1743** is located between the crimping body **171** and the crimping member **1744**, and one end of the elastic member **1743** is connected to the crimping body **171**, and the other end is connected to the crimping member **1744**. For example, the elastic member **1743** may be a spring, and the crimping member **1744** may be a crimping plate.

[0116] In a specific embodiment, the phase shifter **1** may include one phase shift assembly **10**, or may include a plurality of phase shift assemblies **10**. When the phase shifter **1** includes a plurality of phase shift assemblies **10**, the plurality of phase shift assemblies **10** are sequentially disposed along the axial direction of the rotating shaft **12**, in other words, the phase shift assemblies **10** are stacked along the axial direction of the rotating shaft **12**. Stacking the phase shift assemblies **10** helps reduce a volume of the phase shifter **1**.

[0117] FIG. **14** is a diagram of a structure of a phase shifter according to another possible embodiment of this application. As shown in FIG. **14**, in an optional embodiment, the phase shifter **1** includes a plurality of phase shift assemblies **10**, and at least one phase shift assembly **10** and a phase shift assembly **10** adjacent to the at least one phase shift assembly **10** share one drive assembly **14**. Therefore, a quantity of components can be reduced. This facilitates miniaturization and a lightweight property of the phase shifter **1**, and can also reduce costs.

[0118] Specifically, the drive assembly **14** may include the power part and two power output ends **141**. The two power output ends **141** are sequentially disposed along the axial direction of the rotating shaft **12**, and are both in transmission connection to the power part. Two adjacent phase shift assemblies **10** are respectively a first phase shift assembly **10a** and a second phase shift

assembly **10b**. One of the two power output ends **141** is in transmission connection to a drive arm **15** of the first phase shift assembly **10a**, and the other of the two power output ends **141** is in transmission connection to a drive arm **15** of the second phase shift assembly **10b**.

[0119] Further, still refer to FIG. **14**. The phase shift assembly **10** may include a shielding panel **18**, to improve structural strength of the phase shift assembly **10**. Specifically, the substrate **11** is fastened to the shielding panel **18**, and the rotating arm **13** is located on a side that is of the substrate **11** and that faces away from the shielding panel **18**. When the phase shifter **1** includes a plurality of phase shift assemblies **10**, the shielding panel **18** can support a corresponding substrate **11** to enhance structural strength of the phase shifter **1**, and can also reduce or even shield signal interference between two adjacent phase shift assemblies **10**. For example, the shielding panel **18** may be a metal partition plate or a dielectric plate on which a metal layer is formed.

[0120] In addition, the phase shifter **1** provided in this embodiment shields signal interference between two adjacent phase shift assemblies **10** through the shielding panel **18**. Compared with a related technology in which a support plate and a cover plate are disposed for each phase shift assembly **10** to form a separate cavity, a thickness of the phase shifter **1** can be reduced, and a weight of the entire phase shifter **1** can be greatly reduced.

[0121] In a possible embodiment, when the phase shifter **1** includes one phase shift assembly **10**, a metal housing **19** may be disposed on a side that is of a rotating arm **13** and that faces away from a substrate **11**, to shield external signal interference, and improve an electrical indicator of the phase shifter. When the phase shifter **1** includes a plurality of phase shift assemblies **10** sequentially disposed along the axial direction of the rotating shaft **12**, a metal housing **19** may be disposed between two adjacent phase shift assemblies **10**, to shield signal interference between the two adjacent phase shift assemblies **10**, and improve an electrical performance indicator of the phase shifter.

[0122] Specifically, when the phase shifter **1** is disposed in a radome, a mounting clip may be fastened to the radome. The mounting clip includes a clamping part. Clamping parts are connected to phase shift assemblies **10** in one-to-one correspondence, to fasten the phase shift assemblies **10** to the radome.

[0123] It is clear that a person skilled in the art can make various modifications and variations to this application without departing from the protection scope of this application. In this way, this application is also intended to cover these modifications and variations to this application provided that they fall within the scope of the claims of this application and equivalent technologies thereof.

Claims

1. A phase shifter, comprising a phase shift assembly, wherein the phase shift assembly comprises a substrate, a rotating arm, a drive arm, and a drive assembly, wherein an output line is disposed on the substrate, and the output line comprises an arc-shaped segment; and the rotating arm is rotatably connected to the substrate through a rotating shaft, a coupling line is disposed on the rotating arm, and the coupling line is electrically connected to the arc-shaped segment; and one end of the drive arm is fastened to the rotating arm, and the other end extends to a side that is of the rotating shaft and that faces away from the output line, and is in transmission connection to the drive assembly; the drive assembly drives the drive arm to move, and the drive arm drives the rotating arm to rotate relative to the substrate; and in a process in which the rotating arm rotates relative to the substrate, the coupling line slides along the arc-shaped segment.
2. The phase shifter according to claim 1, wherein the rotating arm and the drive arm are an integrated structure.
3. The phase shifter according to claim 1, wherein the drive assembly comprises a power output end, the power output end is in transmission connection to the drive arm, the drive assembly drives the power output end to reciprocate along a first direction, and the power output end drives the

drive arm to drive the rotating arm to rotate.

4. The phase shifter according to claim 3, wherein the first direction is in a same plane as a symmetry axis of the arc-shaped segment, and is perpendicular to the symmetry axis of the arc-shaped segment.

5. The phase shifter according to claim 1, wherein the rotating shaft is located on the symmetry axis of the arc-shaped segment.

6. The phase shifter according to claim 3, wherein a guide trough is disposed on the drive arm, and the guide trough extends along a length direction of the drive arm; and the power output end comprises a push shaft, and the push shaft is inserted into the guide trough, and is driven by the drive assembly to slide relative to the guide trough.

7. The phase shifter according to claim 1, wherein the phase shift assembly comprises an elastic crimping assembly, and the elastic crimping assembly is fastened to the rotating arm, and clamps the substrate and the rotating arm, to apply an abutting force toward the substrate to the rotating arm.

8. The phase shifter according to claim 7, wherein the elastic crimping assembly comprises a crimping body, a crimping foot group, and a transition connecting part, wherein the crimping foot group is located on a side that is of the substrate and that faces away from the rotating arm, and abuts against and is slidably connected to a surface that is of the substrate and that faces away from the rotating arm; the crimping body is fastened to a side that is of the rotating arm and that faces away from the substrate, and an elastic crimping part configured to press the rotating arm toward the substrate is disposed; and the transition connecting part connects the crimping body and the crimping foot group.

9. The phase shifter according to claim 8, wherein the substrate has an arc-shaped edge, and the arc-shaped edge is located on a side that is of the output line and that faces away from the rotating shaft; and the elastic crimping assembly slidably fits with the arc-shaped edge; the crimping foot group comprises at least two crimping feet, the transition connecting part comprises at least two connecting arms, and the crimping feet are spaced apart along an extension direction of the arc-shaped edge; the at least two connecting arms are connected to the at least two crimping feet in one-to-one correspondence; and space between two adjacent connecting arms forms a slot; and an insertion block adapted to the slot is disposed at an end that is of the rotating arm and that is close to the arc-shaped edge, and the insertion block is inserted into the slot.

10. The phase shifter according to claim 8, wherein a hollow area is formed on the crimping body, and the elastic crimping part comprises a spring plate disposed in the hollow area.

11. The phase shifter according to claim 1, wherein two output lines are disposed on the substrate, the two output lines are respectively a first output line and a second output line, a circle center of an arc-shaped segment of the first output line coincides with a circle center of an arc-shaped segment of the second output line, the first output line is farther away from the rotating shaft than the second output line, and both the arc-shaped segment of the first output line and the arc-shaped segment of the second output line are electrically connected to the coupling line.

12. The phase shifter according to claim 1, comprising a main feeder and a coupling part that are disposed on the substrate, wherein both the main feeder and the coupling part are located on a side that is of the output line and that faces the rotating shaft, the main feeder is electrically connected to the coupling part, and an extension direction of the main feeder is parallel to the first direction.

13. The phase shifter according to claim 12, comprising a third output line disposed on the substrate, wherein the third output line is located on the side that is of the output line and that faces the rotating shaft, and is electrically connected to the coupling part; and an extension direction of the third output line is parallel to the first direction.

14. The phase shifter according to claim 1, wherein there are a plurality of phase shift assemblies, and the plurality of phase shift assemblies are sequentially disposed along an axial direction of the rotating shaft.

- 15.** The phase shifter according to claim 14, wherein the phase shift assembly comprises a shielding panel, the substrate is fastened to the shielding panel, and the rotating arm is located on a side that is of the substrate and that faces away from the shielding panel.
- 16.** An antenna, comprising a radiating element and a phase shifter, wherein the radiating element is electrically connected to the phase shifter, and the phase shifter is configured to adjust a feeding phase of the radiating element; wherein the phase shifter comprises a phase shift assembly, wherein the phase shift assembly comprises a substrate, a rotating arm, a drive arm, and a drive assembly, wherein an output line is disposed on the substrate, and the output line comprises an arc-shaped segment; and the rotating arm is rotatably connected to the substrate through a rotating shaft, a coupling line is disposed on the rotating arm, and the coupling line is electrically connected to the arc-shaped segment; and one end of the drive arm is fastened to the rotating arm, and the other end extends to a side that is of the rotating shaft and that faces away from the output line, and is in transmission connection to the drive assembly; the drive assembly drives the drive arm to move, and the drive arm drives the rotating arm to rotate relative to the substrate; and in a process in which the rotating arm rotates relative to the substrate, the coupling line slides along the arc-shaped segment.
- 17.** The phase shifter according to claim 16, wherein the rotating arm and the drive arm are an integrated structure.
- 18.** The phase shifter according to claim 16, wherein the drive assembly comprises a power output end, the power output end is in transmission connection to the drive arm, the drive assembly drives the power output end to reciprocate along a first direction, and the power output end drives the drive arm to drive the rotating arm to rotate.
- 19.** The phase shifter according to claim 18, wherein the first direction is in a same plane as a symmetry axis of the arc-shaped segment, and is perpendicular to the symmetry axis of the arc-shaped segment.
- 20.** The phase shifter according to claim 16, wherein the rotating shaft is located on the symmetry axis of the arc-shaped segment.
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